Rinu Swetha.S
241401081

```
user123@localhost:~/page
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_fifo.c
[user123@localhost page]$
```

```
user123@localhost:~/page — /usr/bin/vim page_fifo.c
#include <stdio.h>

int main(){
        int pages[50], frames[10];
        int n, f, i, j,k, found, pageFaults=0;

        //Input
        printf("Enter number of pages: ");
        scanf("%d", &n);

        printf("Enter page reference string: ");
        for(int i=0; i<n; i++){
                scanf("%d", &pages[i]);
        }

        printf("Enter number of frames: ");
        scanf("%d", &f);

        //Initialize frames
        for(i=0; i<f; i++){
                frames[i] = -1;
        }

        int pos = 0; // FIFO pointer

        printf("\nPage\tFrames\n");

        //FIFO Logic
        for(int i=0; i<n; i++){
                found = 0;

                // check if page already exists
                for(j=0; j<f; j++){
                        if(frames[j] == pages[i]){
                                found = 1;
                                break;
                        }
                }

                //if not found -> page fault
                if(found==0){
                        frames[pos] = pages[i];
                        pos = (pos+1)%f; // circular update
                        pageFaults++;
                }

                //print frames
                printf("%d\t", pages[i]);
                for(k=0; k<f; k++){
                        if(frames[k] != -1)
                                printf("%d ",frames[k]);
                        else
                                printf("-");
                }

                printf("\n");
        }
        printf("\nTotal Page Faults = %d\n", pageFaults);

        return 0;
}
```

```
user123@localhost:~/page

[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_fifo.c
[user123@localhost page]$ gcc page_fifo.c
[user123@localhost page]$ ./a.out
Enter number of pages: 5
Enter page reference string: 1 2 3 4 1
Enter number of frames: 3

Page    Frames
1       1 --
2       1 2 -
3       1 2 3
4       4 2 3
1       4 1 3

Total Page Faults = 5
[user123@localhost page]$
```

Rinu Swetha.S
241401081

```
user123@localhost:~/page

[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_lru.c
[user123@localhost page]$ gcc page_lru.c
[user123@localhost page]$ ./a.out
Enter number of pages in reference string:
17
Enter the refernce string:
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1
Enter number of frames:
3

Total Page Faults: 11
Total Page Hits: 6
[user123@localhost page]$
```

```
user123@localhost:~/page — /usr/bin/vim page_lru.c

#include <stdio.h>

int main(){
        int n, frames;

        printf("Enter number of pages in reference string: \n");
        scanf("%d", &n);

        int pages[n];
        printf("Enter the refernce string: \n");
        for(int i=0; i<n; i++){
                scanf("%d", &pages[i]);
        }

        printf("Enter number of frames: \n");
        scanf("%d", &frames);
        int frame[frames], time[frames];

        for(int i=0; i<frames; i++){
                frame[i] = -1;
                time[i] = 0;
        }

        int pageFaults = 0, pageHits = 0, counter = 0;

        for(int i=0 ;i<n; i++){
                int found=0;

                for(int j=0; j<frames; j++){
                        if(frame[j] == pages[i]){
                                found=1;
                                pageHits++;
                                counter++;
                                time[j] = counter;
                                break;
                        }
                }

                if(!found){
                        int LruIndex = 0;

                        for(int j=1; j<frames; j++){
                                if(time[j] < time[LruIndex]){
                                        LruIndex = j;
                                }
                        }

                        counter++;
                        frame[LruIndex] = pages[i];
                        time[LruIndex] = counter;
                        pageFaults++;
                }
        }

        printf("\nTotal Page Faults: %d\n", pageFaults);
        printf("Total Page Hits: %d\n", pageHits);

        return 0;
}
```